BELLWETHER-3 — Preregistration

Series: BELLWETHER (spatial / measurement nonclassicality)
Experiment ID: BELLWETHER-3 (3rd in the BELLWETHER series; 4th RF quantum flagship overall — after BELLWETHER-1, CHRONO-1, BELLWETHER-2)
Title: Peres–Mermin Magic Square — State-Independent Quantum Contextuality bound into an authorization ProofRecord
Owner: Remnant Fieldworks Inc. (Derek Adam Hone)
Status at lock: design + harness complete, validated in simulation ($\chi = 6.000$ across three input states), hardware job pending
Governing framework: RF Experiment–Pilot–IP Covenant (preregistration-first; honest outcomes; claims narrower than evidence; file-first / publish-after for new matter)

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1. One-sentence statement

We measure the **Peres–Mermin magic-square contextuality functional χ** on two qubits, show it exceeds the noncontextual bound **for every input state we prepare**, and **cryptographically bind the measured χ (and its state-independence) into an authorization ProofRecord** — the first authorization primitive whose physical witness is quantum contextuality, the resource that fuels quantum computational advantage.

2. Why this is world-class — a new resource, not a bigger number

RF's prior flagships all witnessed **nonlocality** or **temporal** nonclassicality. BELLWETHER-3 crosses into a **distinct and deeper physical resource: contextuality.**

RF flagship	State / system	Test	Resource witnessed	Needs entanglement?
BELLWETHER-1	2-qubit Bell^+	CHSH ($S>2$)	bipartite nonlocality	yes
BELLWETHER-2	3-qubit GHZ	Mermin-3 ($\lvert M\rvert >2$)	multipartite nonlocality	yes
CHRONO-1	1 qubit, mid-circuit	Leggett–Garg ($K3>1$)	temporal nonclassicality	no
BELLWETHER-3	**2 qubits + ancillas**	**Peres–Mermin ($\chi>4$)**	**contextuality (state-independent)**	**no**

Three reasons this is a genuine frontier result:

1. **Contextuality is the fuel of quantum advantage.** Contextuality — not entanglement — is the property proven necessary for universal quantum computation via magic-state distillation (Howard et al., Nature 2014). Binding this resource into an authorization record ties RF's ExecutionProof program directly to the deepest known quantum-computational resource.
2. **State-independence.** The Peres–Mermin violation ($\chi = 6 > 4$) holds for **every** input state — pure, mixed, product, or entangled. We demonstrate it across $|00\rangle$, $|{+}{+}\rangle$, and an entangled Bell state. No prior RF witness had this property: it means the authorization certificate is **robust to whatever state a preparation adversary supplies.**
3. **Nonlocality-free.** The witness requires **no entanglement and no spatial separation** — it is a property of the measurement algebra itself. This is a fundamentally different (and harder-to-explain-away) form of nonclassicality than a Bell test, and it is, to our knowledge, **never before bound into a cryptographic authorization record.**

3. Physics and design

The Peres–Mermin square — nine two-qubit Pauli observables in a 3×3 grid; each row and column is a set of three mutually commuting observables:

	col 1	**col 2**	**col 3**	**row product**
row 1	X·I	I·X	X·X	+II
row 2	I·Z	Z·I	Z·Z	+II
row 3	X·Z	Z·X	Y·Y	+II
col product	+II	+II	**−II**	

Five contexts multiply to +II and one (column 3) to −II. No noncontextual hidden-variable model can assign definite ±1 values to all nine observables consistent with all six product constraints (the six products would have to multiply to both +I and −I).

Contextuality functional:

```
χ = ⟨R1⟩ + ⟨R2⟩ + ⟨R3⟩ + ⟨C1⟩ + ⟨C2⟩ − ⟨C3⟩
```

where ⟨Ck⟩ is the expectation of the **product of the three individually- and compatibly-measured outcomes** in that context.

- **Noncontextual (classical) bound:** $\chi \le 4$
- **Quantum prediction (any state):** $\chi = 6$

Compatible measurement gadget. The three observables in a context commute, so they are measured jointly. Each observable's ±1 eigenvalue is read with an ancilla Hadamard-test parity gadget: `ancilla |0⟩ → H → controlled-(P₀ on q0, P₁ on q1) → H → measure`; outcome bit b gives eigenvalue $(-1)^b$. Three observables → three ancillas → three bits/shot; the per-shot context correlator is (−1)^(b0+b1+b2). Because the observables commute, the gadgets commute and implement a genuine simultaneous (non-disturbing, in the ideal limit) measurement.

Simulation validation (pre-lock, noiseless AerSimulator, 40 000 shots/context):

state	⟨R1,R2,R3,C1,C2,C3⟩	χ
|00⟩	[+1,+1,+1,+1,+1,−1]	**6.000**
|++⟩	[+1,+1,+1,+1,+1,−1]	**6.000**
Bell (entangled)	[+1,+1,+1,+1,+1,−1]	**6.000**

χ = 6 for all three states — maximal quantum value, state-independence confirmed.

4. Execution plan

- **Backend:** IBM open-plan hardware (prefer `ibm_fez`, else min pending-jobs).
- **Circuits:** 6 contexts × 3 states = **18 circuits**, each 5 qubits (2 data + 3 ancilla), 3 clbits.
- **Shots:** 1000 per context (18 000 total) — sized to remaining open-plan budget (~a few seconds).
- **Mode:** Batch (open plan forbids Session). **Channel:** `ibm_quantum_platform`.
- **Circuits transpiled** to backend ISA (optimization_level=1) before submission.

5. Outcome criteria (fixed in advance — honest, symmetric)

Reported **per state** and aggregated:

Verdict	Condition
CONTEXTUALITY (state-independent, under assumptions)	$\chi > 4$ and $\chi - 4 > 3\sigma$ for **every** state tested
CONTEXTUALITY (state-dependent)	$\chi > 4$ (3σ) for some but not all states
INCONCLUSIVE / HOLD	$\chi > 4$ but not 3σ-significant
NO VIOLATION	$\chi \leq 4$

The measured hardware χ for each state is reported **exactly as observed**. No re-runs to chase a preferred number.

6. Honesty boundary (non-negotiable — carried into every artifact)

- **Compatibility / non-disturbance loophole OPEN:** the ancilla parity gadgets are not perfectly non-disturbing on real hardware; a determined noncontextual modeller can attribute $\chi > 4$ to measurement disturbance / finite-precision compatibility.
- **Detection / fair-sampling assumption required:** imperfect readout.
- **Single device**, no device-independence is claimed. **Not** certified randomness.

- **State-independence is asserted from theory and demonstrated across the states actually run** — not proven for literally all states.
- **Only defensible claim:** "Observed Peres–Mermin state-independent contextuality violation ($\chi > 4$) on quantum hardware across multiple input states, under compatibility + detection trust assumptions, cryptographically bound into the authorization record."

6a. Disclosed pre-execution harness facts

- Proven WITNESS/BELLWETHER/CHRONO harness fixes are baked in: FIX-1 (backend via min pending_jobs), FIX-2 (transpile to ISA), FIX-3 (robust token loader). No physics impact.
- AerSimulator confirms the exact extraction path and $\chi = 6$ for all three states. Hardware path (SamplerV2, Batch) is identical to prior validated experiments.

7. ProofRecord construction (`bellwether-proofrecord-3.0`)

```
raw_counts_hash  = H( canonical(counts_by_state_and_context) )
calibration_hash = H( canonical(calibration_snapshot) )
witness_hash     = H( canonical(contextuality_witness) )   # contains χ per state +
verdicts
context_nonce    = H( lp(raw_counts_hash) || lp(job_id) || lp(calibration_hash) ||
lp(witness_hash) )
record_hash      = H( canonical(all fields except record_hash) )
```

Tampering with any measured χ breaks `context_nonce`; any covered-field edit breaks `record_hash`. Verified by 10/10 pre-lock mock tests (tamper-evidence over both counts and witness) and by the independent `verify.py`.

8. Commercial purpose & strategic value (Covenant requirement)

IP thesis. BELLWETHER-3 extends the ExecutionProof patent family with a **contextuality-based, state-independent authorization primitive**: a ProofRecord whose physical witness (a) is grounded in the resource that provably powers quantum computational advantage, and (b) certifies **robustness to the input state** — the authorization holds regardless of what preparation an adversary supplies. This is a materially different and broader claim surface than a Bell/GHZ witness (which is entanglement- and preparation-specific).

Candidate RF-100 use. State-robust hardware authorization — an ExecutionProof seed whose validity does not depend on trusting the preparation stage, only the (disclosed) compatibility/detection assumptions. Natural fit for supply-chain / firmware-attestation contexts where the prepared state cannot be fully trusted.

Portfolio position. With BELLWETHER (nonlocality: bipartite → multipartite), CHRONO (temporal), and now BELLWETHER-3 (contextuality), RF's ExecutionProof program spans **the three canonical families of quantum nonclassicality** — a coherent, defensible, and unusually broad experimental IP base built entirely on preregistered, independently-verifiable hardware runs. This breadth is itself a strategic moat: it is the difference between "a clever nonce" and "a systematic physical theory of authorization."

9. IP gate

BELLWETHER-3 is **almost certainly new patentable matter** (contextuality-based, state-independent authorization has, to our knowledge, no prior art). Per the Covenant it is held **local-only** (no GitHub / Zenodo / LinkedIn / RF-100 disclosure) until Derek makes the **file-first** determination. Execution is **not** disclosure; running the job and fetching results is permitted.

To God be the glory. Proof Before Power. Verification Before Execution.